

2) first visit vs every visit

① first-visit: average returns following first-visit

↳ most widely studied

- both converge to $V^{\pi}(s)$ as $N \rightarrow \infty$

- first-visit, each return is independent, identically distributed est.

of $V^{\pi}(s)$ w/ finite variance

↳ law of large numbers, sequence of avg's of ests

converge to expected values

↳ each avg is unbiased est, std of error falls as $1/\sqrt{n}$

where $n = \#$ of returns avg'd

Every-visit = also converges qualitatively

No Difference

② 1 state: s ; 1 action: $a \rightarrow a$ brings agent from $s \rightarrow s$ w/ probability p

non-terminal

1-p probability of going to s^* (terminal state)

$r = +1$ on all transitions

$\gamma = 1$, no discount

Suppose 10 step episode w/ return of 10

every-visit: append return each time, list of sum of rewards each time

if average all in list $\text{mean}(1, 10) \Rightarrow \frac{55}{10} = 5.5$

Step	0	1	2	3	4	5	6	7	8	9	10
state	s	s	s	s	s	s	s	s	s	s	s^*
reward	X	1	1	1	1	1	1	1	1	1	1
prob	p	p	p	p	p	p	p	p	p	p	$1-p$
		↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
		10	9	8	7	6	5	4	3	2	1

first visit $\rightarrow \frac{10}{\text{first visit}} = 10$
return

$$V_{n+1} = V_n + \frac{W_n G_n}{C_n} - V_n \frac{W_n G_n}{C_n}$$

5) ① $V_{n+1} = V_n + \frac{W_n}{C_n} [G_n - V_n]$ New Est. = Old Est. + stepsize [Target - Old Est.]

by def. $V_n = \sum_{k=1}^{n-1} \frac{W_k G_k}{\sum_{k=1}^{n-1} W_k}$

$$C_{n+1} = C_n + W_{n+1}$$

$$C_n = C_{n-1} + W_n$$

$$V_{n+1} = \frac{\sum_{k=1}^n W_k G_k}{\sum_{k=1}^n W_k} \rightarrow \frac{\sum_{k=1}^n W_k G_k}{C_n} \rightarrow \frac{W_n G_n}{C_n} + \frac{\sum_{k=1}^{n-1} W_k G_k}{C_n}$$

$$V_{n+1} = \frac{W_n G_n}{C_n} + \frac{\sum_{k=1}^{n-1} W_k G_k}{C_n} \cdot \frac{C_{n-1}}{C_{n-1}} + V_n \cdot \frac{C_{n-1}}{C_n}$$

$$V_{n+1} = \frac{W_n G_n}{C_n} + V_n \cdot \frac{C_{n-1} + W_n - W_n}{C_{n-1} + W_n} \rightarrow \frac{W_n G_n}{C_n} + V_n \frac{(C_{n-1} + W_n) - V_n W_n}{(C_{n-1} + W_n)}$$

$$V_{n+1} = \frac{W_n G_n}{C_n} + V_n \frac{(C_{n-1} + W_n)}{(C_{n-1} + W_n)} - V_n \frac{W_n}{C_n} \rightarrow \frac{W_n G_n}{C_n} + V_n - \frac{V_n W_n}{C_n}$$

$$\hookrightarrow \frac{W_n}{C_n} + \frac{W_n}{C_n} [C_n - V_n] = V_{n+1}$$

② The off-policy MC control, for estimating $\pi \approx \pi_x$

- it chooses actions greedily $\pi(S_t) \leftarrow \arg \max_a Q(S_t, a)$

⊛ w/ tie breaking for equal values

↳ it creates deterministic policies

$\pi(A_t | S_t) \rightarrow$ prob of choosing A_t given S_t , for det. policy it's always = 1

$$\text{therefore } W = \frac{\pi(A_t | S_t)}{b(A_t | S_t)} = 1$$

Ex 4

- 1) a) $Q_{n+1} \leftarrow Q_n + \frac{1}{n} (R_n - Q_n)$
 $N \leftarrow \text{New} \leftarrow \text{old Est} + \text{step size} (\text{Target} - \text{old Est.})$

$R \leftarrow \text{New} \leftarrow \text{old}$

$$N(A) \leftarrow N(A) + 1$$

$$Q(A) \leftarrow Q(A) + \frac{1}{N(A)} [R - Q(A)]$$

Append G_t to $\text{Returns}(S_t)$

$$V(S_t) \leftarrow \text{average}(\text{Returns}(S_t))$$

new

$$\text{init } N(S_t) = 0 \quad N(S_t) \leftarrow N(S_t) + 1$$

$$V(S_t) \leftarrow V(S_t) + \frac{1}{N(S_t)} [G_t - V(S_t)]$$

if

$$N(S_t, A_t) = 0 \quad \text{Count state-action pair occurrences}$$

$$N(S_t, A_t) \leftarrow N(S_t, A_t) + 1$$

original

$$G_t \leftarrow \gamma G_t + R_{t+1} \quad \text{returns update}$$

add G_t to returns

$$Q_n = \frac{1}{N} \sum_{i=1}^N G_n(S_i, A_i) + (N-1) \left(\sum_{i=1}^{N-1} G_n(S_i, A_i) \cdot \frac{1}{(N-1)} \right)$$

separate new return from previous returns

$$Q_n(S_i, A_i) = Q_{n-1}(S_i, A_i) + \frac{1}{N} (G_n(S_i, A_i) - Q_{n-1}(S_i, A_i))$$

average (Returns(S_t, A_t)) average all returns for $S-A$ pair every time

$$Q_n(S_t, A_t) = Q_{n-1}(S_t, A_t) + \frac{1}{N} (G_n(S_t, A_t) - Q_{n-1}(S_t, A_t))$$

get new Q as before

inside checking loop for first time visit;

$$Q_n(S_t, A_t) \leftarrow Q(S_t, A_t) + (G - Q(S_t, A_t)) \cdot \frac{1}{N(S_t, A_t)}$$

$$\text{then; } \arg \max_a Q(S_t, a) \quad R_t \rightarrow \gamma V(S_t)$$

$$G_n = R + \gamma G_{n-1}$$